Assessment of AIS Antenna Placement near Hall-Effect Thrusters for Small Satellite Applications

Longer mission lifetimes and improved manoeuvrability enabled by electric propulsion on small satellites have introduced complex interactions between radio-frequency (RF) systems and Hall-effect thruster plumes. Dense, partially ionized plumes perturb the local plasma environment, potentially degrading RF payload performance through antenna detuning, phase instability, and altered wave propagation, particularly for low-frequency systems. This study presents a physics-based modelling framework to assess plume–RF interactions affecting VHF Automatic Identification System (AIS) antennas. The approach integrates plume flow-field modelling, plasma parameter estimation, and cold-plasma electromagnetic simulations to quantify changes in plasma frequency, permittivity, and ion density during thruster operation. Flux-based surface accumulation models further evaluate contamination effects from neutralized propellant and charge-exchange ions, revealing gradual antenna detuning due to dielectric film formation. Results indicate that low-frequency RF systems are significantly more susceptible to plume-induced effects than higher-frequency payloads, underscoring the importance of plume–RF coupling analysis in early spacecraft design and motivating mitigation strategies such as increased antenna–thruster separation, operational constraints, and protective coatings.

Keywords - Electric propulsion, Hall-effect thruster plume, Plume–RF interaction, Spacecraft plume contamination, small satellite RF systems

I. INTRODUCTION

Hall-effect thrusters (HETs) are becoming the favoured propulsion option for small satellite missions, because of their high specific impulse, low propellant mass requirements, and long operational lifespan. Despite these advantages, HET operation modifies the spacecraft's surrounding plasma environment by introducing a partially ionised xenon plume. The interaction between ionised plume species and communication electromagnetic fields can become a crucial design constraint when spacecraft integrate sensitive radio-frequency (RF) payloads, especially VHF antennas like Automatic Identification System (AIS) receivers[11.]. RF antennas operating close to the local plasma frequency suffer from significant dispersion, detuning, and signal attenuation. Low-frequency antennas (VHF, UHF) are especially susceptible because their carrier frequencies lie near the plasma cutoff frequencies typical of HET plumes. Conversely, high-frequency antennas (S-band, X-band) are far less impacted because their wavelengths are significantly smaller than the scale of plasma perturbations.

This paper develops a complete plume–RF coupling framework combining:

- Hall thruster plume expansion modelling
- Plasma parameter estimation
- Cold-plasma electromagnetic wave propagation modelling
- Flux-based xenon contamination modelling
- RF frequency sensitivity comparison

The objective is to determine how electric propulsion operation degrades AIS antenna performance and to propose safe integration guidelines for small satellite missions.

II. HALL THRUSTER PLUME MODELLING METHODOLOGY

The behaviour of plasma species in a Hall-effect thruster plume is governed by a combination of quasi-neutral ion–electron dynamics, neutral gas expansion, and charge-exchange (CEX) processes. To evaluate the plume's impact on nearby antennas, it is essential to estimate the spatial variation of ion density, neutral density, ion flux, divergence behaviour, and backflow contamination rates. This section develops a comprehensive flow field model by combining empirical Hall thruster plume laws, analytical formulations, and directly coded numerical simulation.

A. Thruster Operating Parameters[1.]

A representative low-power Hall Effect thruster is considered, consistent with the performance envelope of 40 to 100 W HETs frequently used onboard for small satellites in LEO of 500 kms. A station keeping of +/- 1 km is maintained at 500 km altitude, and the thruster is fired at a station keeping maneuver is performed when the satellite crosses the 1 km altitude decay.

1) Ion Exhaust Velocity

The ion velocity is estimated from beam voltage:

$$v_i = \sqrt{\frac{2q_e V_b}{mX_e}}\ (m/s) \qquad \text{Eq (2.1)}$$

Using a discharge voltage of V_b = 78V, the ion exhaust velocity is 10.74 km/s which is the required range of Exhaust velocity that can be achieved in a Hall Effect Thruster.

2) Ionic beam current and Ionic flux Rate

In the Hall effect thruster, the ions are accelerated by the electric field and forms a beam that carries the electric current, the total number of ions emitted per second is directly tied to the measured beam current, in other words the Ionic flux rate (φ) is directly dependent on the Ionic beam current (I_b)

$$I_b = \frac{\dot{m}}{m} * q\ (Amperes) \qquad \text{Eq (2.2)}$$

$$\varphi = \frac{I_b}{q}\ (ions/second) \qquad \text{Eq (2.3)}$$

3) Divergence Half Angle

The plume exhibits a relatively large beam divergence half-angle of 66° (α), indicating that the accelerated ions spread over a broad angular range downstream of the thruster. Such a wide divergence can affect not only the net thrust efficiency but also the level of ion flux incident on nearby spacecraft surfaces, solar arrays, or sensitive instruments. Characterizing these parameters is therefore essential for understanding the thruster's performance, assessing spacecraft–plume interactions, and optimizing mission designs.

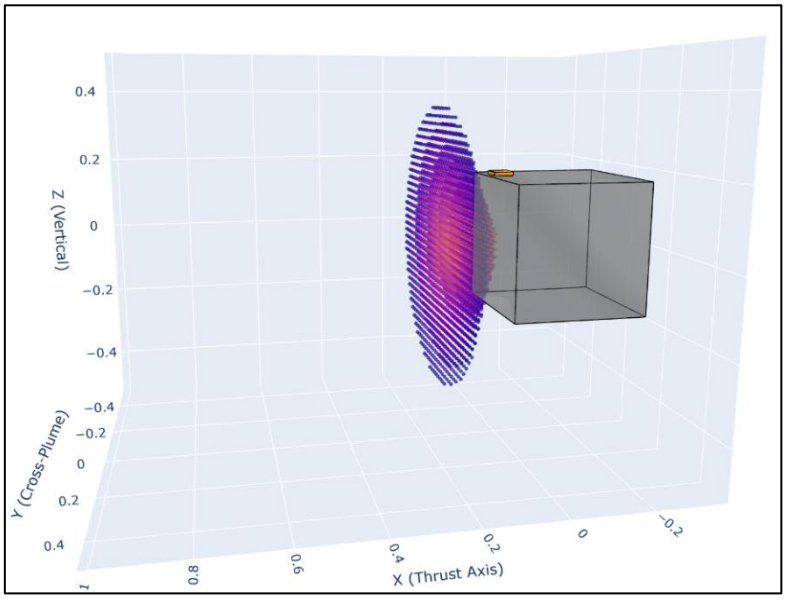


Figure 1: HET Plume for a divergence half angle of 66 degrees

4) Summary of parameters

TABLE I Summary of Thruster operating parameters

Parameter	***Value***
Beam Voltage	78 V
Beam Current	0.48 A
Ion Exhaust velocity	10.7 km/s
Ion Flux	3 x 10^8 ions/s
Divergence half-angle	66 degrees

B. Plume Expansion and Density Field[2.],[9.]

The plume downstream of a Hall thruster expands in a conical pattern governed by ion beam divergence. The model for this analysis follows a simplified analytical framework and assumed to expand within a conical region of the half-angle. All ions are distributed across this cone, and their density decreases with geometric spreading.

5) Total Ion Flux and Plume Geometry

The starting point for modelling the Hall thruster plume is the assumption that ions emerging from the discharge channel expand into space in a conical pattern defined by the plume divergence half-angle α. This is a widely accepted representation for low-power Hall thrusters, where the applied magnetic field and geometry of the channel result in a plume that approximates a spherical sector rather than a narrow beam.

The effective plume area at distance “R” downstream is:

$$A(r) = 2\pi(1 - Cos\ \alpha)R^2 \quad \text{Eq (2.4)}$$

Eq 2.4 ensures the conservation of ion numbers as the plume expands and naturally enforces the $1/r^2$ decay in density due to geometric spreading.

Flux Density of AIS Antenna at an effective distance of R from the thruster is:

$$Flux\ density = \frac{\dot{m}}{a} \quad \text{Eq (2.5)}$$

$$Local\ number\ density\ [n(z)] = \frac{\varphi}{V_e} \quad \text{Eq (2.6)}$$

The corresponding ion number density at a distance r is obtained by distributing the total ion production rate over the plume volume and dividing by the average velocity. This can be helpful is computing density values that decrease as the thruster-antenna distance increases.

6) Maximum ion flux via Gaussian Approximation

The ion concentration near the plume axis is typically higher due to the internal acceleration structure of the Hall thruster. Experimental measurements (IEPC, CNES, JAXA) show that Hall thruster plumes exhibit a **Gaussian-like** angular profile, with the highest density at the centre and progressively decreasing density toward the edges. This Gaussian width approximation is consistent with experimental plume angular distributions reported in IEPC/JAXA measurements.

To approximate this behaviour analytically, the beam width parameter σ is estimated geometrically:

$$\sigma = \frac{L}{\sqrt{3}}\ (meters) \quad \text{Eq (2.7)}$$

$$L = \frac{Exit\ radius}{\tan\ (divergence\ anagle)} \quad \text{Eq (2.8)}$$

The maximum ionic flux follows:

$$n_{max} = \frac{\varphi}{(2 * \pi * \sigma^2)} \quad \text{Eq (2.9)}$$

Where φ is the total ion flux in ions/s obtained using Eq (2.3)

7) Neutral Density Background[7,]

From a practical perspective, not all xenon atoms supplied to a Hall Effect Thruster are ionized. A fraction of propellant remains neutral and expands thermally from the cathode region and the discharge channel, these are called “Fast ions”. These Fast ions are uncharged ions and they neither contribute to the electromagnetic plasma effects nor contribute heavily to the Contamination and CEX (Charge Exchange) processes.

The ions exit from the thruster with much lower velocities than the ions and have a broader spatial distribution. Thus, modelling the neutral density is essential for the estimating the xenon deposition on the antennas.

The neutral density plays a crucial role in determining the Charge-exchange rate because:

- Higher Neutral density slows down the ion production

- Slow Ions are the primary source of surface deposition

CEX rate is proportional to

$$n = \frac{\varphi}{Q}\ (Particles/m^3) \qquad \text{Eq (2.10)}$$

Where, φ is the total ion flux and Q is the volume flow rate at exit of the nozzle

$$Q = \frac{Exhaust\ Velocity}{Area\ of\ Nozzle}\ (m^3/s) \qquad \text{Eq (2.11)}$$

8) Charge-Exchange (CEX) Backflow Modelling[2.],[6.],[9.]

Charge-Exchange (CEX) is a dominant process in low-power Hall thruster plumes; it occurs when an energetic ion collides with a slow neutral atom and exchanges charge:

$$_{fast}Xe^{+} + {}_{slow}Xe \rightarrow {}_{fast}Xe + {}_{slow}Xe^{+}$$

The new ions produced by CEX has:

- Almost zero velocity
- No longer aligned with the plume axis
- Can therefore travel back towards the spacecraft

The phenomenon of these ions travelling back to the spacecraft is called as the "Backflow effect", the backflow effect is the main source of metal sputtering, Ionic contamination, Xenon film deposition, Optical system clouding and even Antenna impedance detuning. The CEX process depends on the Xenon ion-neutral cross section σ_{ex}, and for a typical Hall effect thruster is lies near 1-3 x 10^{19} m^2. The reaction rate is proportional to both the ion density and neutral density[7.]:

$$N_{CEX} \propto \dot{n}_i n_n \sigma_{cx} V_i\ (m^3/s) \qquad \text{Eq (2.12)}$$

The standard consideration for the backflow ion flux is 10%. However, a slightly higher margin of 15% is adopted to provide a conservative margin. The higher margins accounts for uncertainties such as thruster geometry, spacecraft proximity effects and Charge-exchange scattering, any of which could potentially increase the upstream flux beyond nominal levels. The fraction may also increase further depending upon external factors such as operational conditions, local plume surface interactions.

9) Summary of Plume output characteristics

Table II: Plume Output characteristics

Parameter	Value
Plume cone area at R	~ 0.23790 m^2
Flux Density at AIS	1.13E+18 ions/m^2/s
Local number density	1.06E+14 /m^3
Total ion flux	3x10^{18} ions/s
Mass flow rate (Thrust derived)	6.521x10^{-7} kg/s

Mass flow rate (Considering 10% losses)	5.8762×10^{-7} kg/s
Max ion flux	1.28×10^{23} ions/m^2/s
Peak Neutral Density	1.58×10^{18} particles/m^3
Backflow Ion flux	$4.04 - 4.49 \times 10^{17}$ ions/s

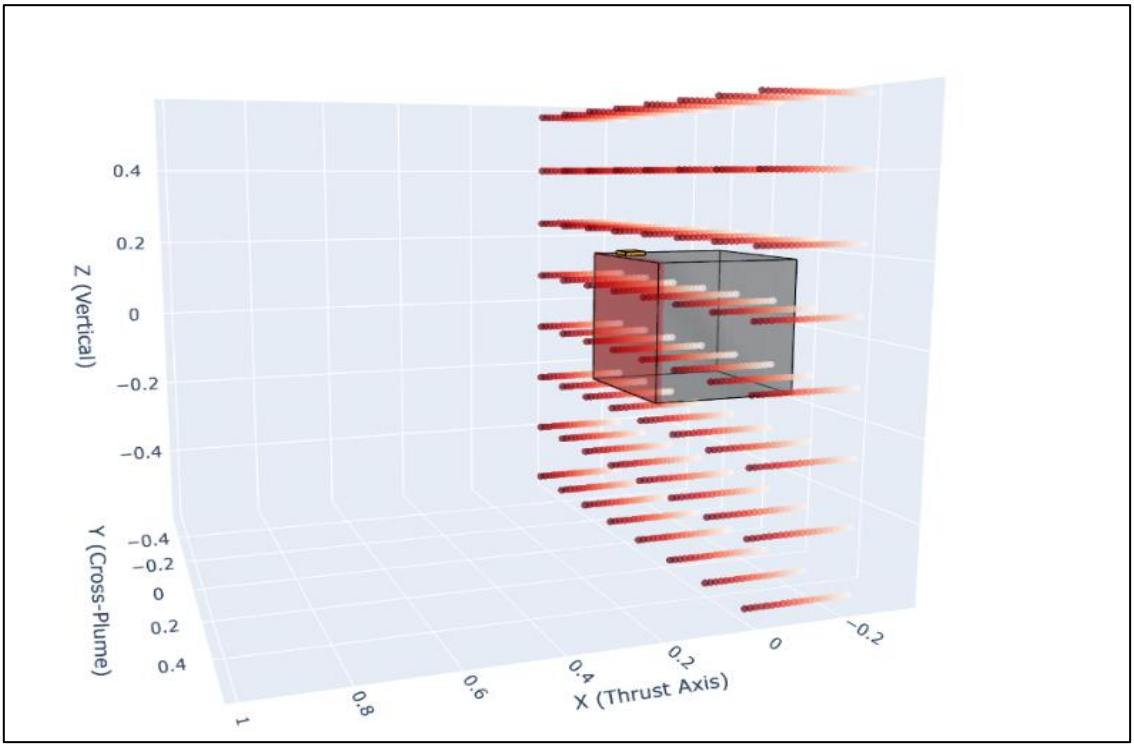


Figure 2: Ionic Backflow for a divergence angle of 66°

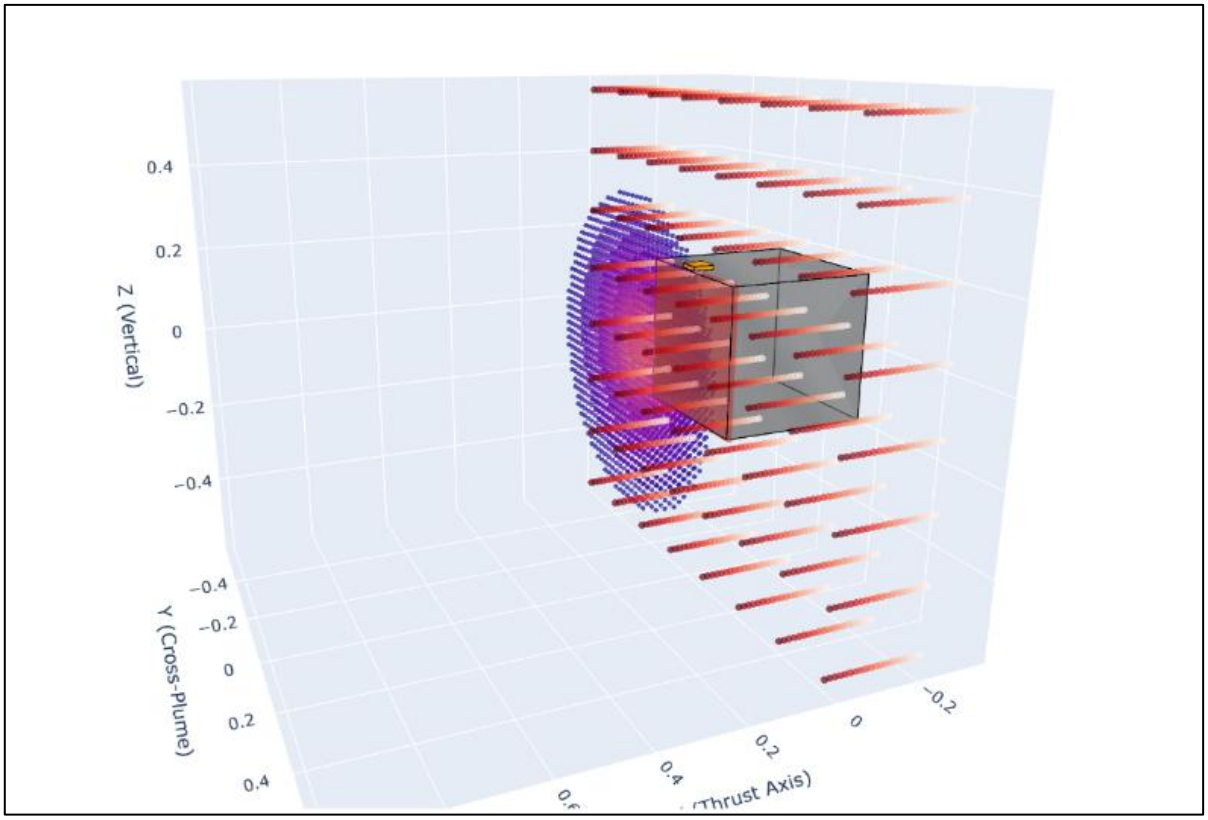


Figure 3: Plume Divergence with ionic backflow

III. PLASMA FREQUENCY AND ELECTROMAGNETIC PROPERTIES OF THE PLUME[9.]

The electromagnetic environment of the ionized plasma is essential for quantifying the effect on VHF AIS antennas. The plasma generated by a Hall-effect thruster consists of ions and electrons, with electrons dominating the radio-frequency response due to their significantly smaller mass. The behaviour of EM waves propagating through such a medium is determined by the electron number density, which directly influences the local plasma frequency, dielectric permittivity, and wave propagation characteristics.

A. Plasma Frequency Derivation[3.], [8.]

The plasma frequency represents the natural oscillation frequency of electrons in response to an applied electromagnetic field.

$$\omega_p = \sqrt{\frac{n_e * e^2}{\varepsilon_o * m_e}}\ (rad/s) \qquad \text{Eq (3.1)}$$

Where,

n_e = Electron number density
e = Electron charge
ε_o = Permittivity of Free space
m_e = Electron Mass

The corresponding linear plasma frequency is:

$$f_p = \frac{\omega_p}{2\pi}(MHz) \qquad \text{Eq (3.2)}$$

The plasma frequency defines the threshold between the propagating and evanescent electromagnetic behaviour. If $f < f_p$ the wave becomes non-propagating and it decays exponentially, and if $f > f_p$ the wave can propagate but with modified phase velocity and wavelength.

Using the local number density derived from flux-based plume model in Section 2.2.1, n(z) ≈ 10^{14} m^{-3} the resulting plasma frequency lies in the VHF band (50-120 MHz). This places the AIS operating frequency (156 – 162 MHz) in the weakly dispersive regime where $f > f_p$, but close enough to cause significant phase distortion and attenuation.

B. Cold Plasma Dielectric Permittivity[8.]

The hall effect thruster plume behaves as a weakly collisional, low temperature plasma. In such a medium, the radio-frequency response is dominated by the collective behaviour of electrons because their mobility is much greater than that of ions. Under the cold-plasma approximation which is valid when the thermal velocities are small compared to wave phase velocities, the relative dielectric permittivity is:

$$\varepsilon_r = 1 - \left(\frac{\omega_p}{\omega}\right)^2 \qquad \text{Eq (3.3)}$$

Where,

ω_p is the plasma angular frequency
$\omega = 2\pi f$ is the RF Signal angular Frequency

For the electron densities obtained in Section 2.2.1 (≈10^{14} m^{-3}), the resulting plasma frequencies fall between 80 and 120 MHz. Since AIS signals operate at 156–162 MHz, the condition $\omega > \omega_p$ is satisfied; however, the ratio is close enough that the permittivity is significantly reduced

$$\varepsilon_r = 0.65 - 0.75$$

A medium with $\varepsilon_r < 1$ alters the propagation in two ways:

The effective refractive index $n_{eff} = \sqrt{\varepsilon_r} < 1$

The impedance seen by the antenna changes, affecting matching and radiation efficiency

This reduction in permittivity is one of the primary phenomena influencing AIS Antenna behaviour when immersed in plume

C. Propagation Constant and Dispersion Relation

Electromagnetic waves travelling through a plasma obey the cold-plasma dispersion relation:

$$k = k_o\sqrt{\varepsilon_r} \quad \text{Eq (3.4)}$$

$$Free\ space\ number = k_o = \frac{\omega}{C} \quad \text{Eq (3.5)}$$

Since $\varepsilon_r < 1$, the propagation constant K is smaller than the free-space value. This leads to two important effects relevant to the AIS Performance.

Wavelength Elongation

The wavelength in the plume becomes

$$\lambda_{plasma} = \frac{2\pi}{k} = \frac{\lambda_0}{\sqrt{\varepsilon_r}} \quad \text{Eq (3.6)}$$

For plume permittivity (0.65 – 0.75):

$$\lambda_{plasma} = 1.15 - 1.25\ \lambda_o$$

The elongation effectively changes the electrical size of the AIS antenna, shifting its resonance and disturbing its impedance match.

Increase in Phase velocity

The phase velocity is given by:

$$V_\phi = \frac{\omega}{k} = \frac{c}{\sqrt{\varepsilon_r}} \quad \text{Eq (3.7)}$$

With εr = 0.7:

$$V_\phi = 1.2C$$

A superluminal phase velocity does not violate causality, but it introduces strong phase dispersion, meaning different components of the AIS signal experience different propagation delays. This results in noticeable phase distortion, addressed in Section 4.4.

D. Attenuation and power loss in Weakly Collisional Plasma

Although the Hall thruster plume is generally weakly collisional, electron–neutral collisions still introduce measurable losses. Incorporating collisional damping, the attenuation coefficient becomes:

$$\alpha = \frac{v}{2c}\left(\frac{\omega_p}{\omega}\right)^2 \qquad \text{Eq (3.8)}$$

Where,

V is the Electron-neutron Collision frequency. Even modest collision rates produce significant attenuation when $\frac{\omega_p}{\omega}$ is not negligible.

The total Power attenuation across a plum of Effective length by Gaussian Estimate L is given by A = 8.686 α L. Which yielded a 26-28 dB attenuation at AIS Frequencies, equivalent to an approximately 95% reduction in received signal amplitude and a severe reduction in SNR (Signal-to-Noise Ratio). This level of attenuation is mission critical for AIS packet reception during Thruster firing.

E. Conditions for Evanescence and Signal Cutoff

The plasma can transition from a propagating to a non-propagating medium when $\omega_p > \omega$, when this phenomenon occurs, the relative permittivity becomes negative i.e. εr <0, and the incident RF wave becomes evanescent, and causes it to decay exponentially rather than propagating it. The characteristic skin depth is obtained by Equation 3.4:

$$\delta = \frac{1}{k} \qquad \text{Eq (3.9)}$$

Small regions of dense plasma, such as local ionization peaks or transient bursts during discharge current oscillations may temporarily exceed the AIS operating frequency, creating localized signal blackout regions.

While sustained cutoff is unlikely at distances typical for antenna placement, short-duration or localized cutoff can occur, contributing to intermittent AIS reception failures observed during electric propulsion burns.

F. Summary of estimated Plume Frequency at distance R

R is the downstream propagation distance at which plume density and flux are evaluated. In this work, R corresponds to the thruster-to-antenna line-of-sight separation used in the electromagnetic and contamination assessment.

Table III: Estimated Plume Frequency at distance R

Parameter	Value
Plume cone area at distance R - A_{cone}	0.23790 m^2
Local number density – n(z)	1.06E+14 m^{-3}
Plume Radius - r	≈ 0.57 m
Plasma Angular Frequency- ω_p	5.8026E+08 rad/s
Plasma Frequency – f_p	92.3508 MHz
Cold plasma permittivity and evanescent decay for 156 MHz to 162 MHz	
Angular frequency - ω	9.80E+08 – 1.02E+09 rad/s
Relative Permittivity - εr	0.64955 - 0.67502
Free Space Number - k_o	3.2673 – 3.3929 /m

Evanescent Decay Constant - k	2.63322 – 2.78619 /m
Skin Depth - δ	37.976 – 35.872 cm
Power Attenuation	26.5 – 28.05 dB

All values are representative results obtained using the plume model described in Section 2. They correspond to a typical evaluation distance R from the thruster exit plane but do not reflect any specific mission geometry.

IV. ELECTROMAGNETIC IMPACT ON AIS ANTENNAS[3.],[4.], [11.]

The interaction between a Hall-effect thruster plume and a VHF Automatic Identification System (AIS) antenna is governed by the electromagnetic properties of the plume plasma and the proximity of the antenna to the ionized flow field. When the thruster is firing, the antenna is immersed in a medium whose effective permittivity, dispersion relation, and attenuation characteristics differ substantially from free-space conditions.

As shown in Section 3, the plume electron density produces plasma frequencies in the range of 80–120 MHz, close to the AIS operational band (156–162 MHz). This places AIS signals in a weakly over dense regime where electromagnetic waves propagate but exhibit strong distortions in wavelength, phase, and amplitude

This section presents a detailed analysis of how the HET plume affects AIS antenna performance through modified wavenumber, dispersion effects, impedance detuning, and signal attenuation.

A. Modification of wavenumber and phase velocity

In Vacuum, an electromagnetic wave propagates with wavenumber:

$$k_o = \frac{\omega}{c} \qquad \text{Eq (4.1)}$$

Inside a cold plasma, the effective number becomes

$$k = k_o\sqrt{\varepsilon_r} \qquad \text{Eq (4.2)}$$

And the relative permittivity becomes:

$$\varepsilon_r = 1 - \left(\frac{\omega_p}{\omega}\right)^2 \qquad \text{Eq (4.3)}$$

Since the Hall Effect thruster plume density yields $\varepsilon_r < 1$, the wavenumber decreases relative to the vacuum case. Physically this means that:

The wavelength increases

The phase velocity exceeds the speed of light in vacuum:

$$V_\varphi = \frac{\omega}{k} = \frac{c}{\sqrt{\varepsilon_r}} \qquad \text{Eq (4.4)}$$

With typical plume parameters

$$V_\emptyset \approx 1.15c \; to \; 1.25\; c$$

Implications for AIS Antennas

Since the wavelength inside the plume becomes

$$\lambda_{plasma} = \frac{\lambda_o}{\sqrt{\varepsilon_r}} \quad \text{Eq (4.5)}$$

The antenna experiences an effective change in electrical length. This results in:

- Detuning of the resonant frequency
- Altered radiation pattern
- Reduced Radiation efficiency

These effects degrade the AIS link even before considering attenuation and phase distortion

B. Phase Distortion and Signal Integrity

Phase distortion arises from the difference between the vacuum wavenumber and plasma-modified wavenumber:

$$\Delta\emptyset = (k_o - k)L \quad \text{Eq (4.6)}$$

Where L is the effective plume thickness between the antenna and free space.

From the Plume analysis simulation and plume density measurements, the value of phase distortion comes around 35° to 37°

Such a large phase shift can be problematic for the AIS antenna as it uses GMSK modulation [Gaussian Minimum Shift Keying – A continuous phase frequency modulation scheme commonly used in AIS] where the signal demodulation depends on the phase continuity and precise frequency discrimination.

Consequences for AIS Communication

Phase distortion can lead to:

- Increased symbol timing errors
- Loss of phase coherency
- Higher Bit Error Rate (BER)
- Partial or full packet dropouts
- Degradation of AIS reception sensitivity
- Even small variations in phase (10–12°) can impact VHF narrowband communication systems. Distortions above 30° significantly compromise signal fidelity.

C. Impedance Mismatch and Antenna Detuning

Appropriate impedance matching between the antenna and the RF front-end is crucial to an AIS antenna's performance. The effective dielectric environment is altered by the surrounding plasma during Hall thruster operation, which lowers the relative permittivity below unity. This modification moves the antenna's resonance frequency away from the nominal AIS band and modifies the antenna's electrical length.

$$Z_{plasma} = Z_o/\sqrt{\varepsilon_r} \quad \text{Eq (4.7)}$$

where $Z_0$is the free-space impedance. A decrease in $\varepsilon_r$increases the effective impedance, leading to higher voltage standing wave ratio (VSWR), increased reflected power, and reduced radiation efficiency. In addition, spatial non-uniformity of the plume causes asymmetric dielectric loading, which can distort the antenna radiation pattern and further degrade link performance.

D. Combined Electromagnetic Effects

The combined effect of several electromagnetic phenomena causes the AIS antenna performance to deteriorate during Hall thruster operation. These include:

- Altered propagation constant because of decreased permittivity.

- Phase distortion brought on by plasma dispersion.
- Impedance mismatch and resonant antenna detuning.
- Notable RF attenuation inside the plume.

Each mechanism lowers communication quality on its own, but when they are present at the same time, they have a compounding impact that significantly lowers the dependability of AIS links. This emphasizes the necessity of viewing plume–RF interactions rather than as discrete occurrences.

E. Frequency Dependence of Plume-RF Interactions

There is a considerable frequency dependence on the intensity of RF interference caused by plumes. Strong dispersion and attenuation are caused by the signal frequency being only slightly higher than the plume plasma frequency for AIS antennas operating in the VHF band (156–162 MHz). Higher-frequency RF systems, on the other hand, such S-band and X-band, function at frequencies that are orders of magnitude higher than the plasma frequency, where the relative permittivity is close to unity.

Because of this, during thruster operation, higher-frequency antennas undergo very little phase distortion, impedance change, or attenuation. Because of this frequency dependency, plume interference has little effect on higher-frequency communication payloads but is a major threat for VHF AIS systems.

V. CONTAMINATION EFFECTS DUE TO CHARGE-EXCHANGE IONS[5.], [9.],[12.]

Hall-effect thruster operation adds long-term degradation processes through plume-induced surface pollution in addition to electromagnetic interactions. Charge-exchange (CEX) ions and slow neutralized propellant atoms, which can travel upstream and deposit on spacecraft surfaces, are the main causes of this contamination. Gradual material deposition can change surface electrical characteristics and cause performance deterioration over the course of a mission when antennas are situated inside or close to the plume backflow-zone.

This part measures the contaminant flux incident on the AIS antenna and assesses how it affects resonance stability and antenna impedance.

A. Source of contaminating species[6.]

When neutral xenon atoms in the plume engage with rapid xenon ions, charge-exchange collisions take place. The resultant ions are no longer limited by the primary plume direction and inherit the low velocity of the neutral atom. These sluggish ions can propagate back toward the spacecraft, forming a population that is roughly isotropic.

The expression for the overall CEX ion production rate is:

$$N_{CEX}=f_{CEX}Ni \qquad \text{Eq (5.1)}$$

f_{CEX} represents the backflow fraction of the ion beams undergoing the charge exchange process. Based on the experimental observations and conservative design practices, a representative value for $f_{CEX} = 0.15$ is adopted into this study.

These slow ions, together with neutral atoms, dominate the contamination environment experienced by the nearby RF antennas.

B. Ion and Neutral Flux incident on Antenna Surfaces

The particle flux incident on an antenna surface at a distance R cab be computed by the local number density as:

$$\emptyset = nv \qquad \text{Eq (5.2)}$$

Where n is the local CEX ion density and v is the corresponding velocity of the particle. For CEX ions, the velocities are significantly lower than the primary beam velocity, increasing the probability of surface interaction and adhesion.

C. Contamination Film Growth

Over cumulative thruster firing time T, the total deposited mass results in a thin surface film whose thickness is given by:

$$t = \frac{\dot{m}_{dep} T}{A\rho} \quad \text{Eq (5.3)}$$

Where ρ is the effective density of the deposited xenon film.

These deposits are enough to change the surface electrical properties of RF antennas, even though the resultant coating thickness is usually on the scale of micrometres each year. AIS antennas are especially sensitive to minute variations in surface impedance and dielectric loading because they function at long wavelengths.

D. Impact of Contamination of AIS Antenna Performance

The deposited xenon layer behaves as a dielectric coating on the conductive antenna surface. This modifies the effective electrical length and shifts the resonant frequency according to:

$$\lambda_{eff} = \frac{\lambda_o}{\sqrt{\varepsilon_{eff}}} \quad \text{Eq (5.4)}$$

Where ε_{eff} is the effective permittivity of the contaminated surface layer.

The primary RF impacts include:

- Resonant frequency drift
- Increased input impedance mismatch
- Elevated VSWR
- Reduction in antenna gain
- Long-term degradation of reception sensitivity

Unlike electromagnetic plume effects discussed in Section 4, contamination effects are cumulative and irreversible, persisting even when the thruster is not firing.

E. Relative Sensitivity of VHF Antennas

Compared to higher-frequency RF systems, low-frequency antennas like AIS are far more susceptible to contamination. Even thin layers can result in detectable detuning because the deposited film thickness represents a higher percentage of the effective wavelength at VHF. On the other hand, for S-band or X-band antennas, the same contamination thickness results in a tiny proportion of the wavelength and has little effect on radio frequency.

The notion that placing AIS antennas close to electric propulsion systems need extra care is supported by this frequency dependency.

F. Summary of Contamination Effects

Plume induced contamination contributes to AIS performance degradation in the following ways:

- Upstream transport of CEX ions and neutrals
- Gradual accumulation of dielectric films on antenna surfaces
- Long-term impedance mismatch and detuning
- Persistent reduction in communication margin

When combined with the previous section that dealt with Electromagnetic effects, further operational strategies should be considered regarding antenna placement for small satellites employing Hall-Effect thrusters.

VI. FREQUENCY SENSITIVITY OF RF SYSTEMS[11.]

The RF system's operating frequency has a significant impact on how severe Hall-effect thruster plume interference is. When the signal frequency is near the local plasma frequency, plume-induced electromagnetic effects become considerable, as demonstrated in Sections 3 and 4. AIS signals (156–162 MHz) are in a weakly over dense and dispersive propagation domain for the plume circumstances examined in this work since the plasma frequency falls within the VHF range.

Significant phase distortion, impedance mismatch, and attenuation are experienced by AIS signals in this range. Higher-frequency RF systems, including S-band and X-band, on the other hand, function at frequencies that are orders of magnitude higher than the frequency of the plume plasma. In these situations, the plume functions as free space and the relative permittivity is close to unity.

Because of this, during thruster operation, higher-frequency antennas show very little phase distortion, little attenuation, and constant impedance characteristics. Furthermore, because the cumulative film thickness only makes up a small portion of the wavelength, contamination-induced detuning effects are greatly less at higher frequencies. This frequency dependency explains why plume–RF interaction has no effect on higher-frequency communication payloads but is a crucial design consideration for VHF AIS antennas.

VII. DISCUSSION

The study's findings show that Hall-effect thruster plumes can seriously impair adjacent VHF AIS antenna performance via surface pollution and electromagnetic interaction. While charge-exchange-induced pollution causes cumulative, long-term degradation, plasma dispersion and attenuation are the main causes of short-term performance loss during thruster firing.

The combined consequences include impedance mismatch, resonant detuning, a significant decrease in received signal strength, and phase distortion that exceeds permissible limits for GMSK-modulated transmissions. When antennas operate concurrently with thruster firing or are situated inside the near-field plume region, these impacts are quite severe.

These results emphasize the significance of assessing plume–RF interaction early in the system planning phase from the standpoint of spacecraft design. Instead of being seen as separate subsystems, antenna positioning, operational sequencing, and propulsion duty cycles must be considered collectively. Low-frequency antennas like AIS need special mitigation techniques to guarantee dependable operation, but higher-frequency RF systems are mainly immune to plume effects.

VIII. CONCLUSION

A physics-based evaluation of Hall-effect thruster plume effects on VHF AIS antennas in small satellite platforms was reported in this work. The study measured both short-term and long-term degradation processes influencing AIS performance by integrating plume flow field modelling, plasma parameter estimates, cold-plasma electromagnetic analysis, and contamination modelling.

The research reveals that, mainly because of plasma dispersion effects, AIS antennas operating close to Hall thrusters are susceptible to considerable phase distortion, attenuation, and impedance mismatch during thruster firing. Furthermore, during a mission, charge-exchange-driven contamination causes a progressive build-up of dielectric layer on antenna surfaces, which results in irreversible detuning. Higher-frequency RF systems, including S-band and X-band, on the other hand, show very little susceptibility to plume-induced effects.

The paper suggests using contamination-resistant design techniques, avoiding simultaneous thruster operation and AIS reception, and ensuring enough antenna–thruster separation to address these problems. These results highlight the need of taking plume–RF coupling into account when designing early spacecraft systems and offer useful integration suggestions for tiny satellites using electric propulsion.

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